

Aalok Patwardhan

✉ aalpatya@gmail.com | 🏠 aalpatya.github.io

Education

PhD — Robotics and Computer Vision | Advisor: Prof. Andrew J. Davison FEng FRS
IMPERIAL COLLEGE LONDON — DYSON ROBOTICS LAB

London, UK
2021 – 2025

- **Distributed Multi-Robot Coordination**
for path planning, information acquisition and consensus within a swarm using Gaussian Belief Propagation (GBP) for Factor Graph inference.
- **Monocular Visual Odometry**
using RGB data for Simultaneous Localisation and Mapping (SLAM), exploiting scene structure for rotation estimation with uncertainty, using dense prediction of surface normals

MEng Engineering — 1st Class with Distinction
UNIVERSITY OF CAMBRIDGE – EMMANUEL COLLEGE

Cambridge, UK
2014 – 2018

- Computer Vision | Signal Processing | (Optimal) Control Systems | Deep Learning

Publications

- **Aalok Patwardhan**, Andrew J. Davison, 2025.
DANCeRS: A Distributed Algorithm for Negotiating Consensus in Robot Swarms. IEEE ICRA 2026.
- **Aalok Patwardhan***, Callum Rhodes*, Gwangbin Bae, Andrew J. Davison, 2025.
U-ARE-ME: Uncertainty-Aware Rotation Estimation in Manhattan Environments. * Equal contribution. 3DV 2025.
- **Aalok Patwardhan**, Andrew J. Davison, 2024.
A Distributed Multi-Robot Framework for Exploration, Information Acquisition and Consensus. IEEE ICRA 2024 (**oral**).
- **Aalok Patwardhan**, Riku Murai, Andrew J. Davison. 2023.
Distributing Collaborative Multi-Robot Planning With Gaussian Belief Propagation. IEEE Robotics and Automation Letters, 8(2): 552-559. (**oral**).
- **Aalok Patwardhan**, Andrew J. Davison, 2023.
Distributed Formation Planning for Robot Swarms. Workshop on Distributed Graph Algorithms for Robotics at IEEE ICRA 2023.

Relevant Experience

Peer Reviewer, IEEE — ROBOTICS & AUTOMATION LETTERS (RA-L) | CONFERENCES: ICRA, IROS

Lead Research Scientist, Computer Vision — CARTWHEEL

2025 - now

- Leading development of the company's core video-to-3D animation system, spanning model design, system integration, and deployment.
- Designing and integrating a pipeline combining pose estimation, camera tracking, and SLAM for reconstructing 3D motion from monocular video
- Training generative deep learning models for human pose estimation and camera estimation from monocular video.

Autonomy Flight Core Intern — SKYDIO INC.

2025

- Trained deep learning models for optical flow prediction using pytorch and internal tools, including profiling and optimization for deployment on an embedded Qualcomm QCU.
- Developed a visual servoing pipeline using real-time optical flow on embedded hardware, integrating tightly with planning and control for drone gimbal stabilisation.
- Integrated perception modules into real-time planning and control pipelines for autonomous flight.

Director of Studies, Engineering and AI — CAMBRIDGE PROGRAMMES LTD.

2023 – now

- Designed and delivered a two-week summer school curriculum on Engineering and AI for pre-university students.
- Managed a team of teachers and mentors; oversaw content delivery, hands-on workshops, and project supervision.

Signal Processing Engineer — CAMBRIDGE CONSULTANTS LTD.

2018 – 2021

- Lead engineer retrofitting generative deep learning models for use in the recovery of lossy compressed audio data.
- System modelling and unit testing of the PHY layer in 4G/5G communications software.
- Researched and deployed low-power signal processing algorithms on embedded systems, improved efficiency by 40%.
- Designed and implemented the sensory pipeline for a crowd-navigating robot including radar data and computer vision.
- Project manager for a portfolio of worth over £200,000 of software license tools, with regular presentations to senior officers at budget planning meetings

Undergraduate Research Placement — UNIVERSITY OF CAMBRIDGE, TOYOTA MOTORS EUROPE

2017

- Modelled the effects of predictive control in semi-autonomous vehicles to minimise discomfort in humans using the principle behind ‘why we can’t tickle ourselves’. Liaised with Toyota for validation against experimental results.

Teaching Experience

2022 – 2025 **Teaching Assistant, Robotics**, Imperial College London

2020 **Presentation Skills Coach**, Cambridge Consultants

2017 – now **Private Tutor**, GCSE, A-Level Mathematics and Physics, University Admissions Guidance

2015 – now **Lead Mentor**, Cambridge Programmes Ltd. Summer School

Invited Talks

2024 **Oxford Robotics Institute**, Oxford University, UK

2024 **Delft University of Technology**, Delft

2024 **TNO Research**, The Hague, Netherlands

Awards

2021 – 2025 **Dyson Research Fellowship Award**, Dyson Ltd. & EPSRC

2023 **Best Poster Prize (runner-up)**, Imperial College London PhD Competition

2018 **Wallace Prize in Engineering**, Emmanuel College, University of Cambridge

Skills

SOFTWARE DEVELOPMENT

- Python, Numpy and PyTorch for deep learning.
- Parallelised C++ for distributed algorithms.
- 3D simulation and OpenGL graphics.
- Gitlab for Continuous Integration (CI).

LANGUAGES

- Fluent in English, Marathi and Hindi.
- Conversational in Spanish and French.